

GENERAL NOTES

1. GENERAL NOTES ON SHEET 2

3		
2		
1		
NO.	TITLE	DRAWING NO.

REFERENCE PLANS

SPECIAL NOTATIONS

PRELIMINARY DOCUMENT ISSUED TO SUPPORT DRAFT RFP. DOCUMENT IS UNDER DEVELOPMENT, AND NOT APPROVED.

RESERVATIONS:

1. POWER DEMAND OF BOW THRUSTER IS RESERVED PENDING BOW THRUSTER SIZING AND PERFORMANCE ANALYSIS.

STINFO DESIGNATION

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	UNITED STATES COAST GUARD		WASHINGTON, D.C. 20593	
	OFFICE OF NAVAL ENGINEERING			
69 FT	HOMELAND SECURITY CUTTER LIGHT ELECTRIC PLANT LOAD ANALYSIS (EPLA)			HSCL
	SIZE	DATE	USCG DRAWING NO.	REV
	A	3/11/2026	69-HSCL-300-001	0

GENERAL NOTES	
1.	INDIVIDUAL CONNECTED LOAD: FULL LOAD INPUT POWER (MEASURED IN KILOWATTS) OF ONE UNIT OF EQUIPMENT.
2.	STANDARD SERVICE LIFE ALLOWANCE IS 10%. DESIGN MARGIN IS 10%.
3.	ANALYSIS UTILIZES NOTIONAL COMPONENTS WHERE THE EXACT LOAD WAS UNKNOWN
4.	MULTIPLE OPERATING CONDITIONS, LISTED ON SHEET 7, WERE ANALYZED AND SUMMARIZED ON SHEET 5.
5.	TOTAL CONNECTED LOAD: THE PRODUCT OF INDIVIDUAL CONNECTED LOAD AND QUANTITY OF UNITS
6.	THE ELECTRIC PLANT STATUS SUMMARY SHEETS DEMONSTRATE THE ELECTRIC PLANT CAPABILITIES INCLUDING SERVICE LIFE ALLOWANCE MARGIN (SLA).
7.	ALL CONDITIONS SHOWN ON EPLA ARE FOR MANNED CONDITIONS. THERE ARE NO UNMANNED CONDITIONS SHOWN ON THIS CALCULATION.
8.	THE FOLLOWING LOADS ARE CONSIDERED TO BE EMERGENCY LOADS: 1. EMERGENCY LIGHTS 2. NAVIGATION LIGHTS 3. FIRE DETECTION AND FIXED FIREFIGHTING SYSTEMS 4. COMMUNICATION EQUIPMENT 5. NAVIGATION EQUIPMENT 6. GENERAL ALARM SYSTEM

REVISIONS					
REV	ITEM	SWBS	DESCRIPTION	DATE	INITIAL
0	-	-	PRELIMINARY RELEASE FOR DRAFT RFP	3/11/2026	-

Electric Plant Status Summary:	
Total Shore Power Capacity (kW) (Single Cord)	83.14
Number of Ship Service Generators	2.00
Generator Rating, (kW) each	65.00
Total Installed Generator Capacity, (kW) each	130.00
Total Installed Capacity, (kW)	130.00

AC Load Summary - Summer						
SWBS	SWBS Load Category	Conn. Load (kW)	SHORE (kW)	TRANSIT (kW)	ATON (kW)	
200	Propulsion Plant	9.88	6.88	0.58	0.58	
300	Electric Plant	74.12	14.89	10.43	21.89	
400	Command & Surveillance	3.79	0.76	1.32	2.37	
500	Auxiliary Systems	43.15	8.50	10.42	24.59	
512	Heating Ventilation & Air Cond (less heaters)	16.48	14.83	14.83	14.83	
512H	HVAC Heaters	30.21	0.00	0.00	0.00	
600	Outfitting	8.93	0.62	0.79	0.79	
583	Deck Equipment	0.00	0.00	0.00	0.00	
Total AC Load		186.57	46.47	38.37	65.05	
Total AC Load w/ 20% Margin +SLA		223.89	55.76	46.04	78.06	
Shore Power Online		-	1.00	0.00	0.00	
Number of Generators Online		-	0.00	1.00	1.00	
Number of Inverters Online		-	0.00	0.00	0.00	
Excess Electrical Plant Capacity		-	36.67	26.63	-0.05	
Generator Loading		-	56%	59%	100%	
Excess Electrical Plant Capacity w/ 20% Margin + SLA		-	27.38	18.96	-13.06	
Generator Loading w/ 20% Margin + SLA		-	67%	71%	120%	

AC Load Summary - Winter						
SWBS	SWBS Load Category	Conn. Load (kW)	SHORE (19°F) (kW)	TRANSIT (-20°F) (kW)	ATON (0°F) (kW)	ICEBREAKING (-20°F) (kW)
200	Propulsion Plant	9.88	6.88	0.58	0.58	0.58
300	Electric Plant	74.12	14.89	10.43	21.89	10.43
400	Command & Surveillance	3.79	0.61	1.27	2.37	2.23
500	Auxiliary Systems	43.15	8.50	10.42	23.38	10.36
512	Heating Ventilation & Air Cond (less heaters)	16.48	1.27	1.27	1.27	1.27
512H	HVAC Heaters	30.21	10.17	27.19	21.48	27.19
600	Outfitting	8.93	1.88	2.05	2.05	2.05
583	Deck Equipment	0.00	0.00	0.00	0.00	0.00
Total AC Load		186.57	44.19	53.21	73.02	54.11
Total AC Load w/ 20% Margin + SLA		223.89	53.02	63.85	87.63	64.93
Shore Power Online		-	1.00	0.00	0.00	0.00
Number of Generators Online		-	0.00	1.00	2.00	1.00
Number of Inverters Online		-	0.00	0.00	0.00	0.00
Excess Electrical Plant Capacity		-	38.95	11.79	56.98	10.89
Generator Loading		-	53%	82%	56%	83%
Excess Electrical Plant Capacity w/ 20% Margin + SLA		-	30.12	1.15	42.37	0.07
Generator Loading w/ 20% Margin +SLA		-	64%	98%	67%	100%

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HOMELAND SECURITY CUTTER LIGHT
ELECTRIC PLANT LOAD ANALYSIS (EPLA)
AC SUMMARY

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Electric Plant Status Summary:	
Total Shore Power Capacity (kW) (Single Cord)	
Number of UPS	2.00
UPS Voltage (VDC)	24.00
NAV Equipment Voltage (VDC)	12.00
UPS Rating, (kW) each	5.00
Total Installed UPS Capacity, (kW) each	10.00

DC Load Summary - Summer

SWBS	SWBS Load Category	Conn. Load (kW)	SHORE (kW)	TRANSIT (kW)	ATON (kW)	
200	Propulsion Plant	1.44	0.29	0.86	1.15	
300	Electric Plant	1.06	0.47	0.66	0.66	
400	Command & Surveillance	1.17	0.21	0.44	0.54	
500	Auxiliary Systems	9.36	1.98	0.26	3.80	
512	Heating Ventilation & Air Cond (less heaters)	0.46	0.28	0.28	0.28	
512H	HVAC Heaters	0.00	0.00	0.00	0.00	
600	Outfitting	0.65	0.00	0.00	0.00	
583	Deck Equipment	0.00	0.00	0.00	0.00	
Total DC Load		14.14	3.22	2.50	6.43	
Total DC Load w/ 20% Margin + SLA		16.97	3.87	3.00	7.72	
Shore Power Online		-	0.00	0.00	0.00	
Number of UPS Online		-	1.00	1.00	2.00	
Excess UPS Capacity		-	1.78	2.50	3.57	
UPS Loading		-	64%	50%	64%	
Excess UPS Capacity w/ 20% Margin + SLA		-	1.13	2.00	2.28	
UPS Loading w/ 20% Margin + SLA		-	77%	60%	77%	

DC Load Summary - Winter

SWBS	SWBS Load Category	Conn. Load (kW)	SHORE (19°F) (kW)	TRANSIT (-20°F) (kW)	ATON (0°F) (kW)	ICEBREAKING (-20°F) (kW)
200	Propulsion Plant	1.44	0.29	0.86	1.15	0.86
300	Electric Plant	1.06	0.47	0.66	0.66	0.66
400	Command & Surveillance	1.17	0.21	0.44	0.54	0.54
500	Auxiliary Systems	9.36	1.98	0.22	3.80	0.22
512	Heating Ventilation & Air Cond (less heaters)	0.46	0.24	0.24	0.24	0.24
512H	HVAC Heaters	0.00	0.00	0.00	0.00	0.00
600	Outfitting	0.65	0.00	0.00	0.00	0.00
583	Deck Equipment	0.00	0.00	0.00	0.00	0.00
Total DC Load		14.14	3.19	2.42	6.40	2.53
Total DC Load w/ 20% Margin + SLA		16.97	3.82	2.91	7.67	3.03
Shore Power Online		-	0.00	0.00	0.00	0.00
Number of UPS Online		-	1.00	1.00	2.00	1.00
Excess UPS Capacity		-	1.81	2.58	3.60	2.47
UPS Loading		-	64%	48%	64%	51%
Excess UPS Capacity w/ 20% Margin + SLA		-	1.18	2.09	2.33	1.97
UPS Loading w/ 20% Margin + SLA		-	76%	58%	77%	61%

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DC SUMMARY

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